



Operations Research

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Solution to the Training Exercises, Sheet 8

Training Exercises

Exercise T8.1. *Matroids*

Problem

Consider two matroids $U_1 = (E_1, \mathcal{I}_1)$ and $U_2 = (E_2, \mathcal{I}_2)$.

- a) Show that the family of sets $U_1 \cap U_2 := (E_1 \cap E_2, \mathcal{I}_1 \cap \mathcal{I}_2)$ is an independence system ("Unabhängigkeitssystem").
- b) Refute that in general the set system $U_1 \cap U_2 := (E_1 \cap E_2, \mathcal{I}_1 \cap \mathcal{I}_2)$ is a matroid.

Solution

- a) Since U_1 and U_2 are matroids, E_1 and E_2 are finite sets with which $E_1 \cap E_2$ is also finite. Furthermore, $\emptyset \in \mathcal{I}_1$ and $\emptyset \in \mathcal{I}_2$, so $\emptyset \in \mathcal{I}_1 \cap \mathcal{I}_2$ follows. Let $B \in \mathcal{I}_1 \cap \mathcal{I}_2$ and $A \subseteq B$. From $B \in \mathcal{I}_1 \cap \mathcal{I}_2$ follows $B \in \mathcal{I}_1$ and $B \in \mathcal{I}_2$. Since U_1 and U_2 are matroids, it follows that $A \in \mathcal{I}_1$ and $A \in \mathcal{I}_2$, hence $A \in \mathcal{I}_1 \cap \mathcal{I}_2$.
- b) The independence system $U_1 \cap U_2$ is in general not a matroid, which can be observed by considering the following example:

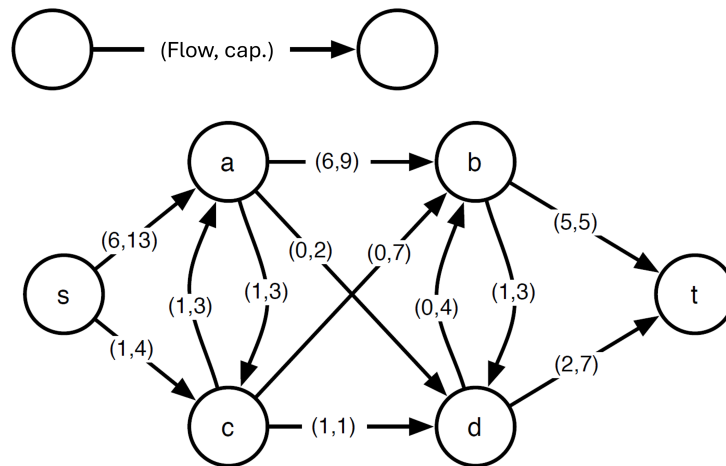
$$\begin{aligned}U_1 &= (\{a, b, c\}, \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}\}) \\U_2 &= (\{a, b, c\}, \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, c\}, \{b, c\}\}) \\U_1 \cap U_2 &= (\{a, b, c\}, \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, c\}\})\end{aligned}$$

The independence system $U_1 \cap U_2$ is no matroid as the exchange property for the sets $\{a, c\}$ and $\{b\}$ is not satisfied.

Exercise T8.2. *Maximum Flow*

Problem

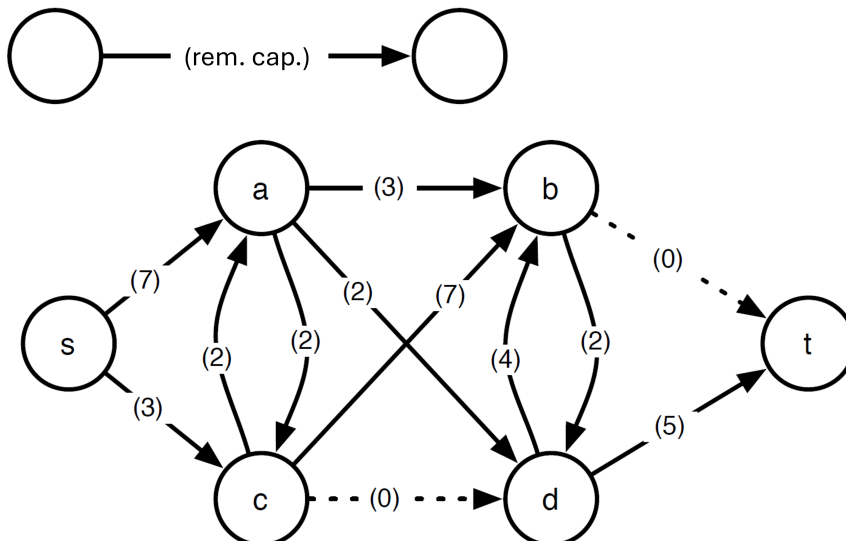
Consider the following network with flow (from s to t) and capacities as indicated.



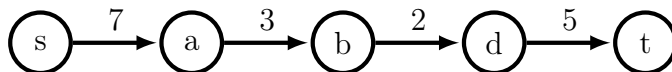
- State the current flow value.
- Based on the current flow, perform two more flow augmentations using the Ford-Fulkerson algorithm. Specify the two associated paths, as well as the resulting flow.
- Is the obtained flow maximal? Give reasons for your answer. What is the maximum flow value here?
- Specify a minimum s, t cut.

Solution

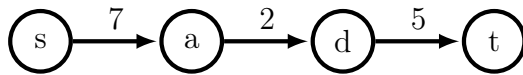
- The value of this flow is 7.
- Given is a network where an existing s, t -flow is already defined. We draw the corresponding residual network (here (as an exception) without flow-induced backward edges):



For example, one can see the following two paths that can be maximally augmented independently:

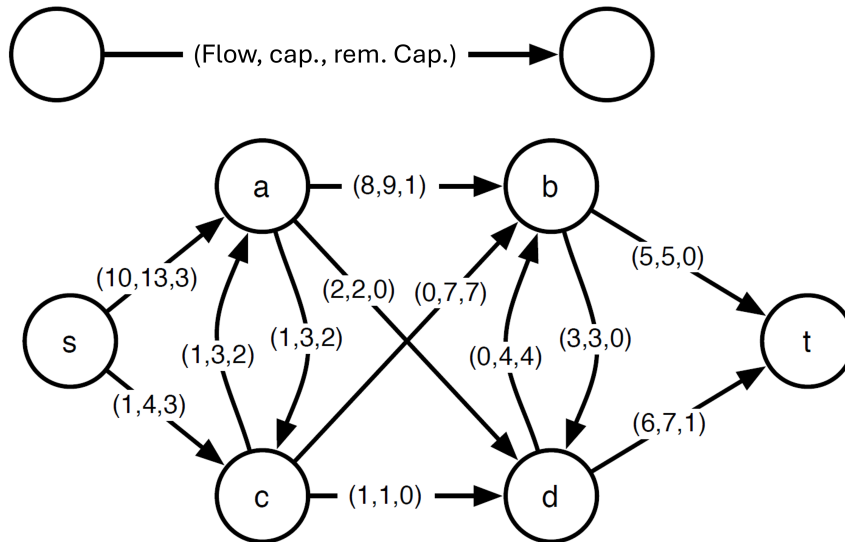


and



(There are also other variants, for example over c , but they always contain the edges (a, d) and (b, d)).

The result of the flow increases for the two paths mentioned above is as follows:



- c) The obtained flow of 11 is maximal. The reason for this is that there is now no s - t path where each edge has a positive residual capacity.
- d) Let $X = \{s, a, b, c\}$ and thus $V \setminus X = \{t, d\}$. The removed edges of the intersection would be $\{(b, t), (b, d), (a, d), (c, d)\}$.

Exercise T8.3. Dredgers

Problem

A construction company has 5 dredgers and needs one of them at each of their 4 construction sites. The distances between the dredgers and the construction sites are given as follows.

dredge	site			
	A	B	C	D
1	90	75	75	80
2	35	85	55	65
3	125	95	90	105
4	45	110	95	115
5	55	85	60	85

Minimize the total transport distance using the Hungarian method. Then specify the optimal assignment and the total transport distance.

Solution

- i) Quadratic matrix

The cost matrix contains more rows than columns. We add a dummy column (cost: 0):

90	75	75	80	0
35	85	55	65	0
125	95	90	105	0
45	110	95	115	0
55	85	60	85	0

ii) Subtract line minima

No effect because each line contains a 0.

iii) Subtract column minima

We subtract the column minima from each column:

55	0	20	15	0
0	10	0	0	0
90	20	35	40	0
10	35	40	50	0
20	10	5	20	0
(-35)	(-75)	(-55)	(-65)	(-0)

iv) Check minimum coverage

Three lines are needed to cover all zeros.

55	0	20	15	0
0	10	0	0	0
90	20	35	40	0
10	35	40	50	0
20	10	5	20	0

v) Add more zeros

The lines needed for line coverage is less than 5. The smallest uncovered number is 5, so we subtract this number from all uncovered elements and add it to all double-covered elements:

55	0	20	15	5
0	10	0	0	5
85	15	30	35	0
5	30	35	45	0
15	5	0	15	0

vi) Check minimum coverage

Four lines are needed to cover all zeros.

55	0	20	15	5
0	10	0	0	5
85	15	30	35	0
5	30	35	45	0
15	5	0	15	0

vii) Add more zeros

The lines needed for line coverage is less than 5. The smallest uncovered number is 5, so we subtract this number from all uncovered elements and add it to all double-covered elements:

55	0	20	15	10
0	10	0	0	10
80	10	25	30	0
0	25	30	40	0
15	5	0	15	5

viii) Check minimum coverage

Five lines are needed to cover all zeros.

55	0	20	15	10
0	10	0	0	10
80	10	25	30	0
0	25	30	40	0
15	5	0	15	5

ix) Optimal allocation

Since we need five lines for minimum coverage, our solution is optimal. The zeros represent the optimal allocation:

55	0	20	15	10
0	10	0	0	10
80	10	25	30	0
0	25	30	40	0
15	5	0	15	5

This corresponds in our cost matrix:

90	75	75	80
35	85	55	65
125	95	90	105
45	110	95	115
55	85	60	85

The assignment of the dredgers to the construction sites is therefore as follows: Dredge 1 ↔ site B, Dredge 2 ↔ site D, Dredge 3 remains unmoved, Dredge 4 ↔ site A and

Dredge 5 $\leftrightarrow$ site C. The total transportation distance is 245 km.

Self-study Exercises

Exercise S8.1. TU and dual of a Matroid

Problem

- a) Given the matrix $A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$. Determine whether A is totally unimodular.

Justify your answer.

- b) Let $M = (E, I)$ be a matroid. We denote by $\mathcal{B}(M)$ the set of bases of M . Now we define the concept of the **dual** of a matroid.

The dual $M^* = (E, I^*)$ of a matroid $M = (E, I)$ contains the same set E as the original matroid. The independent sets are formed as follows: $I^* = \{Z \subseteq E : \exists B \in \mathcal{B}(M) \text{ with } Z \cap B = \emptyset\}$.

(Note: The bases of a matroid are its maximal independent sets.)

- (i) Consider the matroid $M_1 = (E_1, I_1)$ with the set $E_1 = \{a, b, c, d\}$ and $I_1 = \{\emptyset, \{a\}\}$.

State the dual's family of sets I_1^* .

- (ii) Show: The dual $M^* = (E, I^*)$ of a matroid M is an *independence system*.

Solution

- a) Since the first column of A contains three ones, the sufficient condition from the lecture cannot be directly satisfied.

Alternative 1: We show using the sufficient condition that A^T (and thus also A) is totally unimodular.

- $A^T = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$ is totally unimodular because the following conditions are satisfied for $M_1 = \{1, 2\}$ and $M_2 = \{3\}$:

- $a_{ij} \in \{-1, 0, 1\}$ for all i, j .
- Each column contains at most two non-zero coefficients.
- There exists a partition (M_1, M_2) of the set M of rows, such that the following statements are fulfilled:
 - * If two non-zero entries in a column of A have the same sign, then one row is in M_1 and the other in M_2 .
 - * If two non-zero entries in a column of A have different signs, then both rows are either in M_1 or in M_2 .

From the lecture, it is also known that if A is totally unimodular, then A^T is also totally unimodular. From A^T being TU, it follows that A is TU as well.

Alternative 2: We show that every square submatrix of A has a determinant with the value $-1, 0, 1$.

To do this, we consider the square submatrices of size 1×1 , 2×2 , and 3×3 one by one.

- 1×1 : $a_{ij} \in \{-1, 0, 1\}$ for all i, j .
- 2×2 : As soon as a 2×2 submatrix contains a 0 (and all other entries are -1, 0, or 1), the corresponding determinant takes a value in $\{-1, 0, 1\}$. Therefore, it is sufficient to consider only those 2×2 matrices that do not contain zeros. This is the case for only one submatrix:

$$\det \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} = -1 - (-1) = 0 \in \{-1, 0, 1\}$$

- 3×3 :

$$\det(A) = 1 \cdot \det \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - (-1) \cdot \det \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + 0 \cdot \det \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} = 0 \in \{-1, 0, 1\}$$

- Consequently, A is TU.

b) (i) $I_1^* = \{\emptyset, \{b\}, \{c\}, \{d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{b, c, d\}\}$

- (ii) We have $\emptyset \in I^*$, because $\emptyset \subseteq E$ and $\emptyset \cap B = \emptyset$ for an arbitrary basis $B \in \mathcal{B}(M)$. Let $Y \in I^*$ and $X \subseteq Y$. By the definition of the dual, $Y \subseteq E$ and there exists a basis $\tilde{B} \in \mathcal{B}(M)$ such that $Y \cap \tilde{B} = \emptyset$. Trivially, $X \subseteq E$. Additionally, the intersection of X with the previously chosen basis $\tilde{B}$ is necessarily empty. Therefore, $X \in I^*$ and $M^* = (E, I^*)$ is an independence system.

Exercise S8.2. DGS Algorithm

Problem

Consider the following valuation matrix, where bidders are considered to have *unit-demand* valuations, namely each bidder is interested in winning at most a single item in this auction.

	A	B	C
Bidder 1	5	8	3
Bidder 2	9	9	1
Bidder 3	6	2	4

Compute the minimal competitive equilibrium prices and corresponding assignment using the DGS algorithm. Provide all intermediate steps.

Solution

We know that, for the case of assignment markets, a minimum competitive equilibrium is produced as the outcome of the DGS auction. We implement the auction and report the assignment and price vector.

$$\begin{array}{ccc}
A & B & C \\
\begin{pmatrix} 5 & 8 & 3 \\ 9 & 9 & 1 \\ 6 & 2 & 4 \end{pmatrix} &
\begin{pmatrix} 4 & 7 & 3 \\ 8 & 8 & 1 \\ 5 & 1 & 4 \end{pmatrix} &
\begin{pmatrix} 3 & 6 & 3 \\ 7 & 7 & 1 \\ 4 & 0 & 4 \end{pmatrix}
\end{array}$$

In the first round, item set $\{A, B\}$ is overdemanded. The prices for both items are thus raised by 1 to $p = (1, 1, 0)$. In the next round, the set is still overdemanded and prices are once again raised by 1 to $p^* = (2, 2, 0)$. Now, no item is overdemanded and the optimal allocation can be determined as: $\mu(1) = B, \mu(2) = A, \mu(3) = C$.

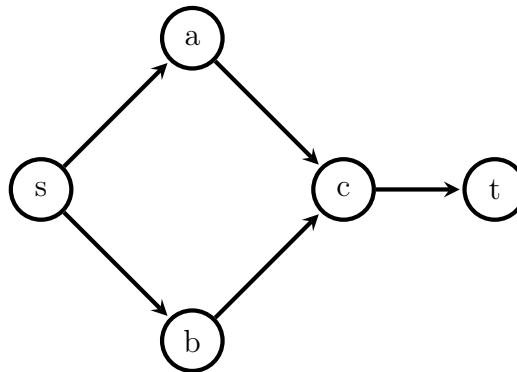
Exercise S8.3. *Maximum flow and minimum cut counterexamples*

Problem

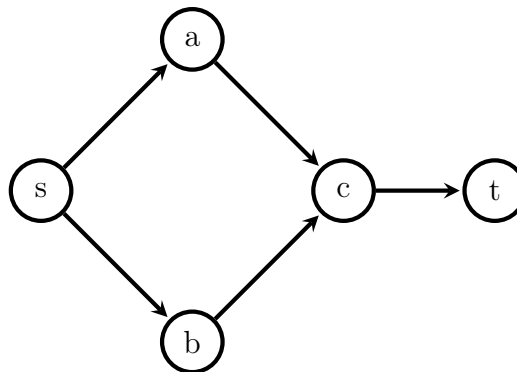
Let $(G = (V, E), s, t, c)$ be a network with source s , sink t , and integral capacities $c(e) \in \mathbb{N}$ for all edges $e \in E$.

Consider the following statements. Use the given graphs to find a suitable counterexample for each one of them. Clearly describe why the given counterexample refutes the statement.

- a) If all capacities are odd then there is a maximal $(s - t)$ -flow f such that $f(e)$ is odd for all $e \in E$.

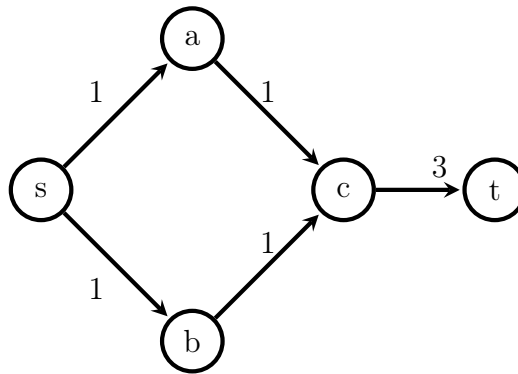


- b) Adding a number $\lambda \in \mathbb{N}$ to all capacities $c(e)$ does not change the minimal cuts.



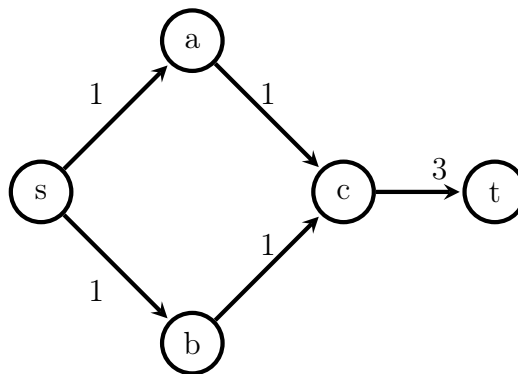
Solution

- a) Consider the following graph:



All capacities are odd. The maximum $s - t$ -flow has a value of 2, which flows over $e := (c, t)$. The flow $f(e) = 2$ is an even number. Therefore, even though all capacities are odd integer numbers, the maximum flow has a flow over edge e which is an even number, showing that the statement is false in general.

b) Consider the following graph:



A minimal cut of value 2 is given by $(\{s\}, \{a, b, c, t\})$. After adding $\lambda = 2$ to all capacities, the minimal value of 5 is achieved by the cut $(\{t\}, \{s, a, b, c\})$, while the value of the initial cut $(\{s\}, \{a, b, c, t\})$ is 6. Therefore, the set of minimal cuts has changed, which shows that the statement is false.